OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 27 OF 55

# Retrieval-Augmented AI System for Intelligent Question Answering from Academic Documents

**Domain:** Retrieval-Augmented Generation & EdTech | **Module:** project_27_academic_qa_rag

## OFFICIAL ABSTRACT

Academic institutions and students work with a large volume of digital resources such as textbooks, lecture notes, research papers, question banks, laboratory manuals, and project documents. Finding specific information from lengthy academic documents can be time-consuming, while conventional keyword-based search systems often fail to understand the actual meaning or context of a user's question. The proposed Retrieval-Augmented AI System for Intelligent Question Answering from Academic Documents aims to provide an intelligent solution that allows users to ask questions in natural language and receive answers based specifically on the information available in their academic documents. The proposed system is implemented using Python, Natural Language Processing, text embeddings, vector database technology, and Large Language Models (LLMs). Uploaded documents are first processed, cleaned, and divided into meaningful text segments. These segments are converted into numerical vector representations using embedding techniques and stored in a vector database. When a user asks a question, the system searches the stored document representations and retrieves the most semantically relevant content before generating the final answer. The major uniqueness of the proposed system is its retrieval-first approach. Unlike a general AI chatbot that generates responses mainly from its pre-trained knowledge, this system grounds its responses in the user's selected academic documents. This allows the system to provide answers that are more relevant to the actual learning material. The semantic retrieval mechanism can also identify related information even when the words used in the user's question are different from the exact wording present in the document. Another important feature is source-aware answer generation. The system can identify the document or relevant section from which the answer was derived, allowing students to verify the generated response against the original academic material. This improves transparency and reduces the risk of users accepting an AI-generated answer without understanding its source. The system can also support follow-up questions while maintaining the

context of the ongoing academic discussion. The proposed system can be useful for students, faculty members, educational institutions, and academic researchers. Students can use it to clarify doubts, locate information from lengthy study materials, and support examination preparation, while faculty members can use it to quickly access information from subject-specific documents. New documents can also be added to the system without retraining the underlying language model, making the solution flexible and scalable for different subjects and academic environments. Overall, the project demonstrates how Retrieval-Augmented Generation (RAG) can transform static academic documents into an interactive and context-aware knowledge system. By combining semantic document retrieval, vector-based search, and AI-generated responses with source references, the proposed system provides a practical approach to intelligent academic question answering while keeping responses closely connected to the actual documents provided by the user.

## 📌 Project Summary & System Specifications

| | |
|---|---|
| **Project Title** | Retrieval-Augmented AI System for Intelligent Question Answering from Academic Documents |
| **Specialization / Category** | Retrieval-Augmented Generation & EdTech |
| **Target Technology Stack** | Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI |
| **Default Backend Port** | `:5027` |
| **Local Implementation** | `project_27_academic_qa_rag` |